

Zero-Shot Classification with CLIP

Deep Learning

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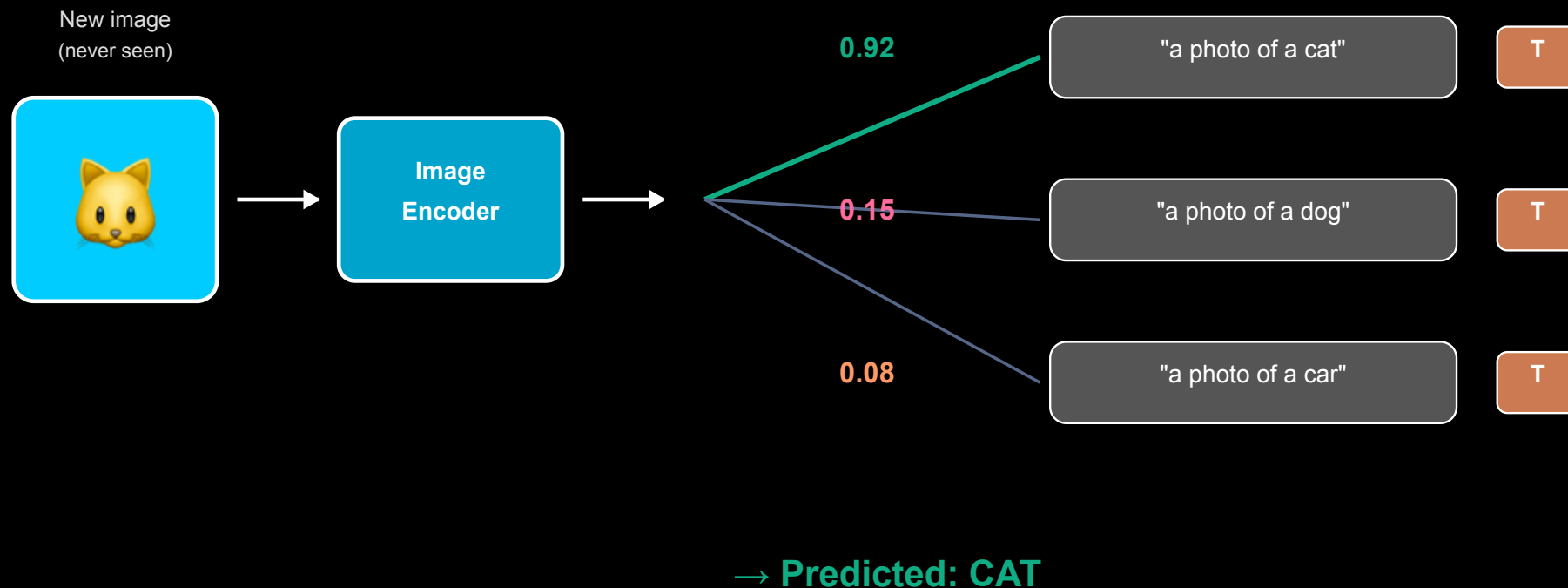
Contents of This Video

In this video, we will cover:

- What is zero-shot classification?
- How CLIP enables zero-shot
- Creating classifier from text prompts
- Zero-shot vs. traditional classification
- Future trends and open questions in deep learning

Zero-Shot Classification

Zero-Shot: Compare Image to Text Descriptions



The Process

1. Define classes with text prompts:

```
"a photo of a cat"  
"a photo of a dog"  
"a photo of a bird"
```

2. Encode all prompts with text encoder:

$$T_1, T_2, T_3 = \text{TextEncoder}(\text{prompts})$$

3. Encode test image:

$$I = \text{ImageEncoder}(\text{image})$$

4. Pick class with highest similarity:

$$\text{class} = \arg \max_i (I \cdot T_i)$$

Prompt Engineering

Simple: “a photo of a cat”

Better: “a photo of a cat, a type of pet”

Best (for specific dataset): Customize to domain

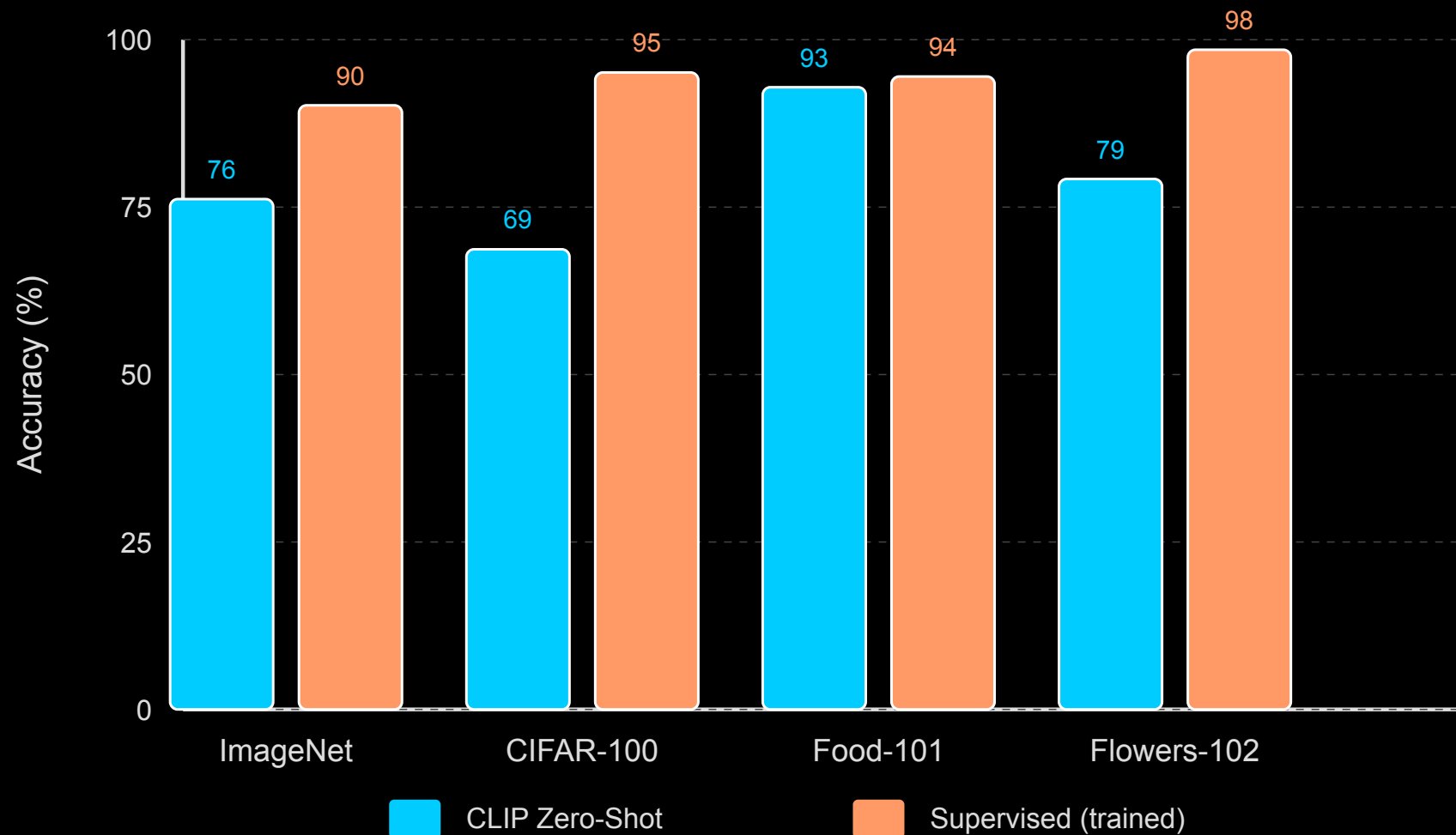
Template variations:

- “a photo of a {class}”
- “a picture of a {class}”
- “an image of a {class}”

Averaging over templates often helps!

Zero-Shot Performance

Zero-Shot vs. Supervised Accuracy



Why This Matters

Traditional:

- Collect labeled data for each new task
- Train or fine-tune a classifier
- Expensive and time-consuming

Zero-Shot:

- Just describe what you're looking for
- Works instantly
- Infinitely flexible class definitions

Applications:

- Image search (“find beach photos”)
- Content moderation
- Rapid prototyping

Future Trends in Deep Learning

Scaling continues:

- Larger models, more data, more compute
- Emergent capabilities appear at scale

Multimodal AI:

- Text + images + audio + video in one model
- Universal representations across modalities

Efficient architectures:

- Mixture of Experts (MoE): activate only relevant parts
- State Space Models (Mamba): alternatives to attention
- Distillation: compress large models into smaller ones

Open Questions in Deep Learning

Why does it work so well?

- Overparameterized yet generalizes
- Double descent phenomenon
- Lottery ticket hypothesis

Fundamental challenges:

- Reasoning vs. pattern matching
- Hallucinations and reliability
- Catastrophic forgetting in continual learning

Societal questions:

- Energy consumption and sustainability
- Bias, fairness, and alignment
- Economic disruption and access

What We've Covered

In this video, we've learned:

- Zero-shot: classify without training examples
- CLIP enables this through shared image-text space
- Describe classes with text prompts
- Prompt engineering improves results
- Future: multimodal AI, efficient architectures, scaling
- Open questions: reasoning, reliability, societal impact